

3. Feature importance via absolute coefficients

For linear-type models, larger $|\text{coefficient}| \rightarrow$ more influence (given similar scale).

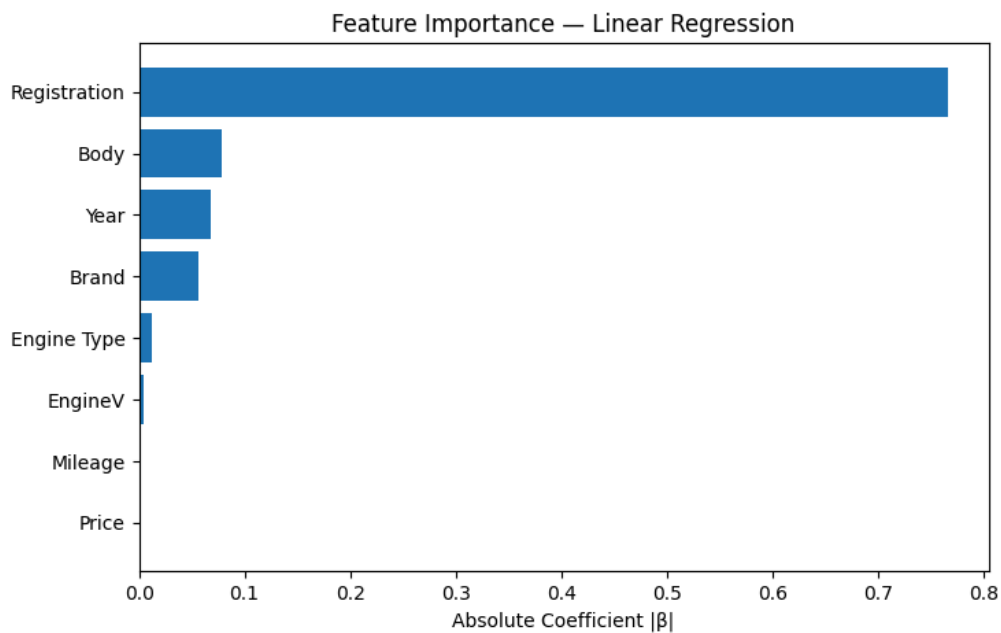
```
coef_df['Linear_Abs'] = coef_df['Linear_Coeff'].abs()  
coef_df['Ridge_Abs'] = coef_df['Ridge_Coeff'].abs()  
coef_df['Lasso_Abs'] = coef_df['Lasso_Coeff'].abs()
```

```
coef_df.sort_values('Linear_Abs', ascending=False)
```

	Feature	Linear_Coeff	Ridge_Coeff	Lasso_Coeff	Linear_%_Change	Linear_Abs	Ridge_Abs	Lasso_Abs
6	Registration	0.766631	0.763506	0.000000	115.250167	0.766631	0.763506	0.000000
2	Body	-0.078491	-0.078477	-0.039125	-7.548956	0.078491	0.078477	0.039125
7	Year	0.067574	0.067593	0.064858	6.990894	0.067574	0.067593	0.064858
0	Brand	-0.056075	-0.055999	-0.013741	-5.453174	0.056075	0.055999	0.013741
5	Engine Type	-0.011897	-0.011807	0.000000	-1.182617	0.011897	0.011807	0.000000
4	EngineV	0.004920	0.004915	0.000000	0.493238	0.004920	0.004915	0.000000
3	Mileage	0.000208	0.000208	-0.000121	0.020789	0.000208	0.000208	0.000121
1	Price	0.000017	0.000017	0.000019	0.001668	0.000017	0.000017	0.000019

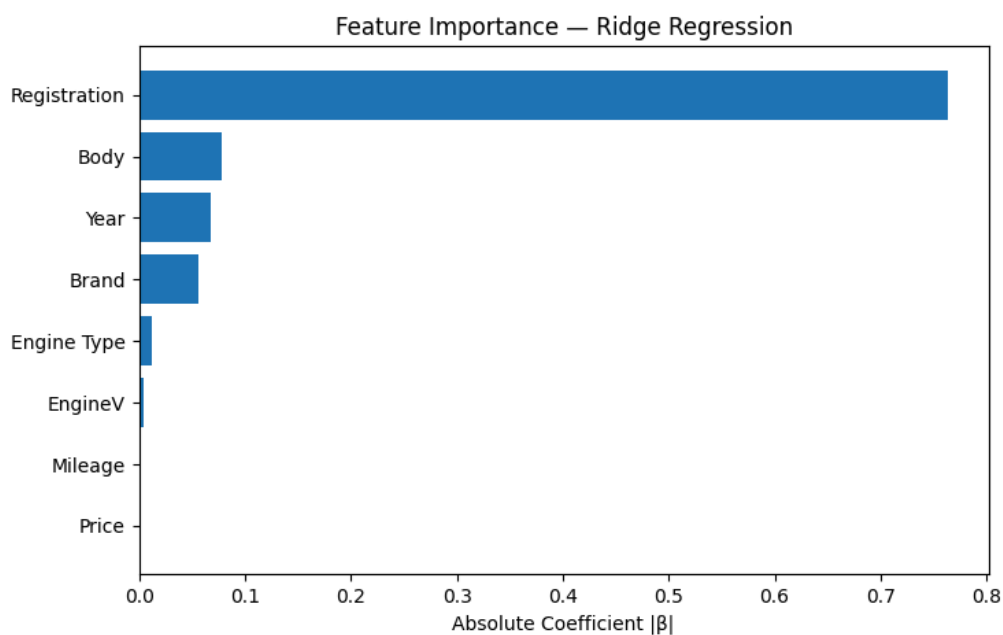
```
plt.figure(figsize=(8,5))
coef_df_sorted = coef_df.sort_values('Linear_Abs', ascending=False)

plt.barh(coef_df_sorted['Feature'], coef_df_sorted['Linear_Abs'])
plt.xlabel('Absolute Coefficient  $|\beta|$ ')
plt.title('Feature Importance - Linear Regression')
plt.gca().invert_yaxis()
plt.show()
```



```
plt.figure(figsize=(8,5))
coef_df_sorted = coef_df.sort_values('Ridge_Abs', ascending=False)

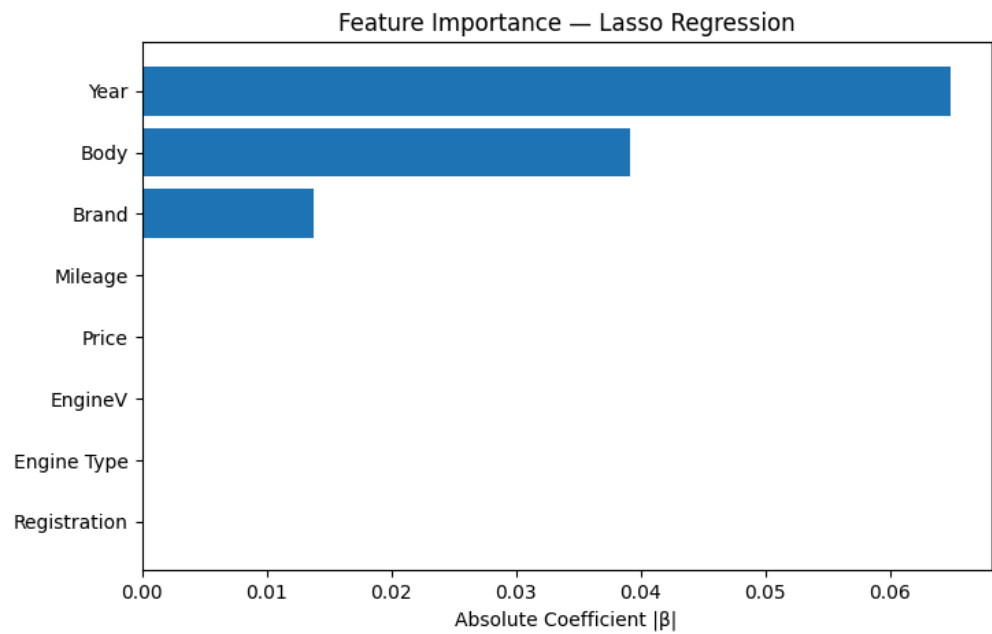
plt.barh(coef_df_sorted['Feature'], coef_df_sorted['Ridge_Abs'])
plt.xlabel('Absolute Coefficient  $|\beta|$ ')
plt.title('Feature Importance - Ridge Regression')
plt.gca().invert_yaxis()
plt.show()
```



```
plt.figure(figsize=(8,5))
coef_df_sorted = coef_df.sort_values('Lasso_Abs', ascending=False)

plt.barh(coef_df_sorted['Feature'], coef_df_sorted['Lasso_Abs'])
plt.xlabel('Absolute Coefficient  $|\beta|$ ')
plt.title('Feature Importance - Lasso Regression')
plt.gca().invert_yaxis()
plt.show()
```

plt.show()



4. What Ridge vs Lasso tell you

- Ridge: Coefficients are shrunk but rarely exactly zero.

Interpretation: all features contribute a bit; large ones are more important.

- Lasso: Some coefficients may be exactly 0.

Interpretation: Lasso thinks those features are not useful and effectively drops them → feature selection.

```
# You can see which features Lasso removed:
coef_df[coef_df['Lasso_Coeff'] == 0]
```

	Feature	Linear_Coeff	Ridge_Coeff	Lasso_Coeff	Linear_%_Change	Linear_Abs	Ridge_Abs	Lasso_Abs
4	EngineV	0.004920	0.004915	0.0	0.493238	0.004920	0.004915	0.0
5	Engine Type	-0.011897	-0.011807	0.0	-1.182617	0.011897	0.011807	0.0
6	Registration	0.766631	0.763506	0.0	115.250167	0.766631	0.763506	0.0

```
# And which it kept:
coef_df[coef_df['Lasso_Coeff'] != 0].sort_values('Lasso_Abs', ascending=False)
```

	Feature	Linear_Coeff	Ridge_Coeff	Lasso_Coeff	Linear_%_Change	Linear_Abs	Ridge_Abs	Lasso_Abs
7	Year	0.067574	0.067593	0.064858	6.990894	0.067574	0.067593	0.064858
2	Body	-0.078491	-0.078477	-0.039125	-7.548956	0.078491	0.078477	0.039125
0	Brand	-0.056075	-0.055999	-0.013741	-5.453174	0.056075	0.055999	0.013741
3	Mileage	0.000208	0.000208	-0.000121	0.020789	0.000208	0.000208	0.000121
1	Price	0.000017	0.000017	0.000019	0.001668	0.000017	0.000017	0.000019